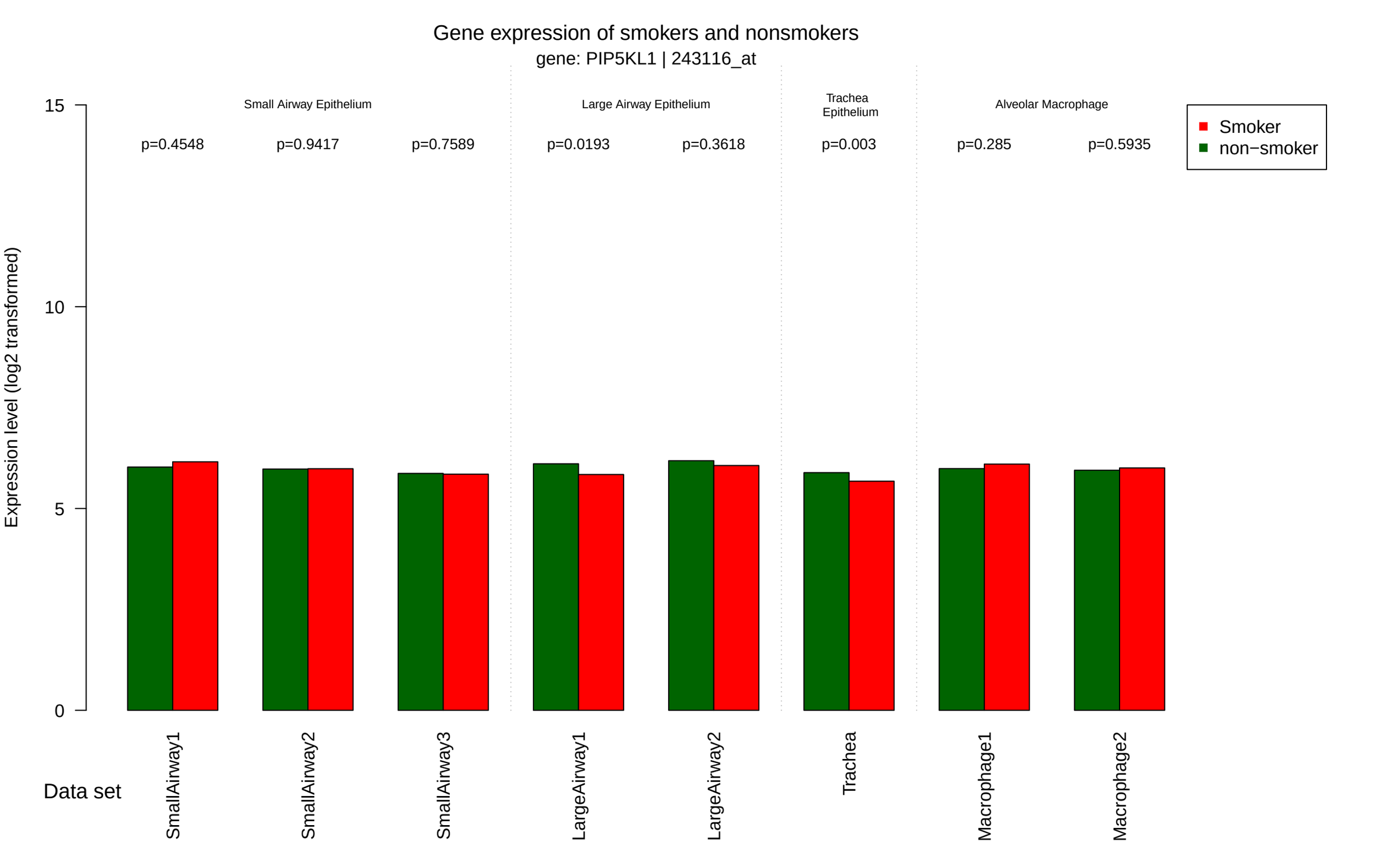
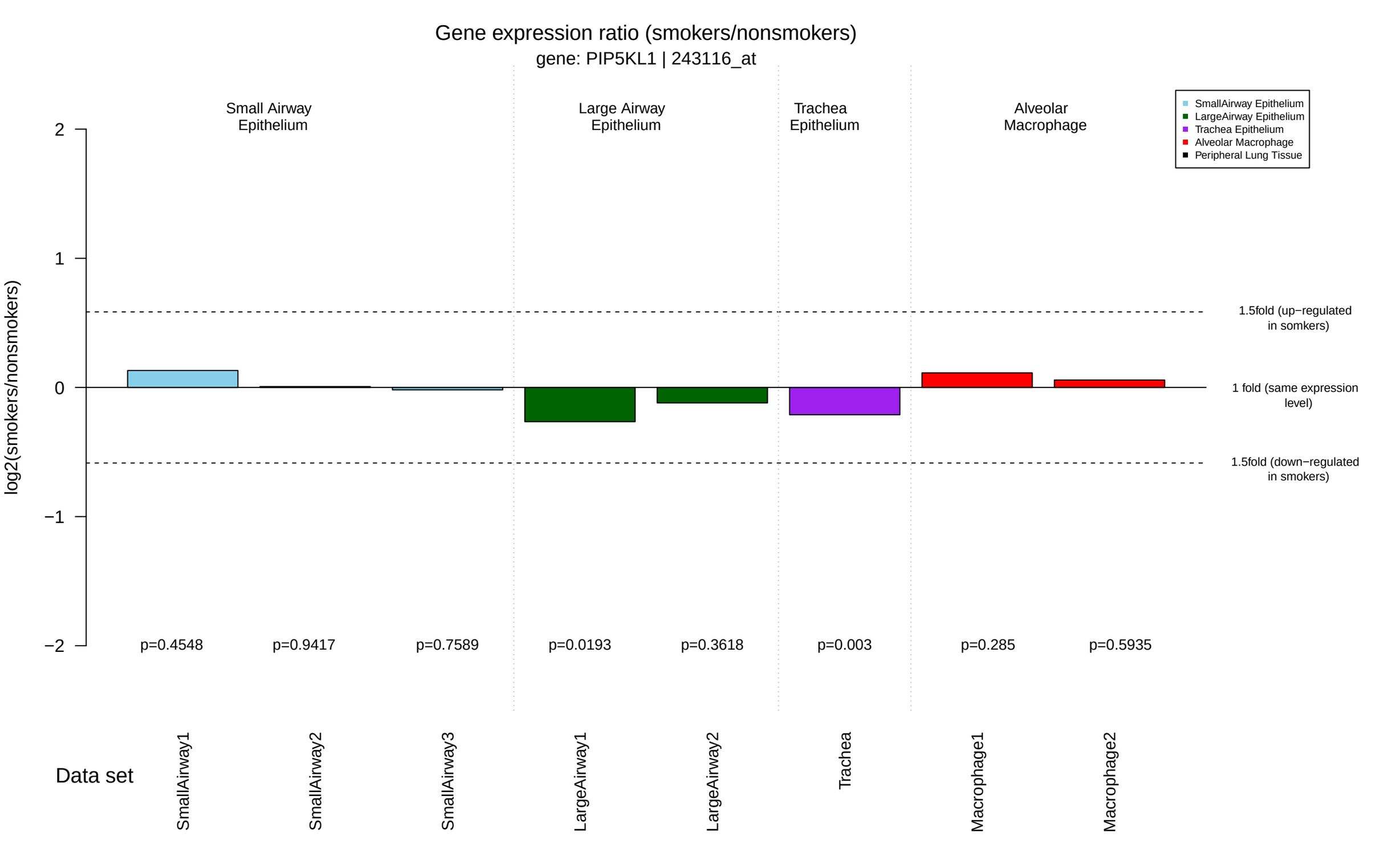
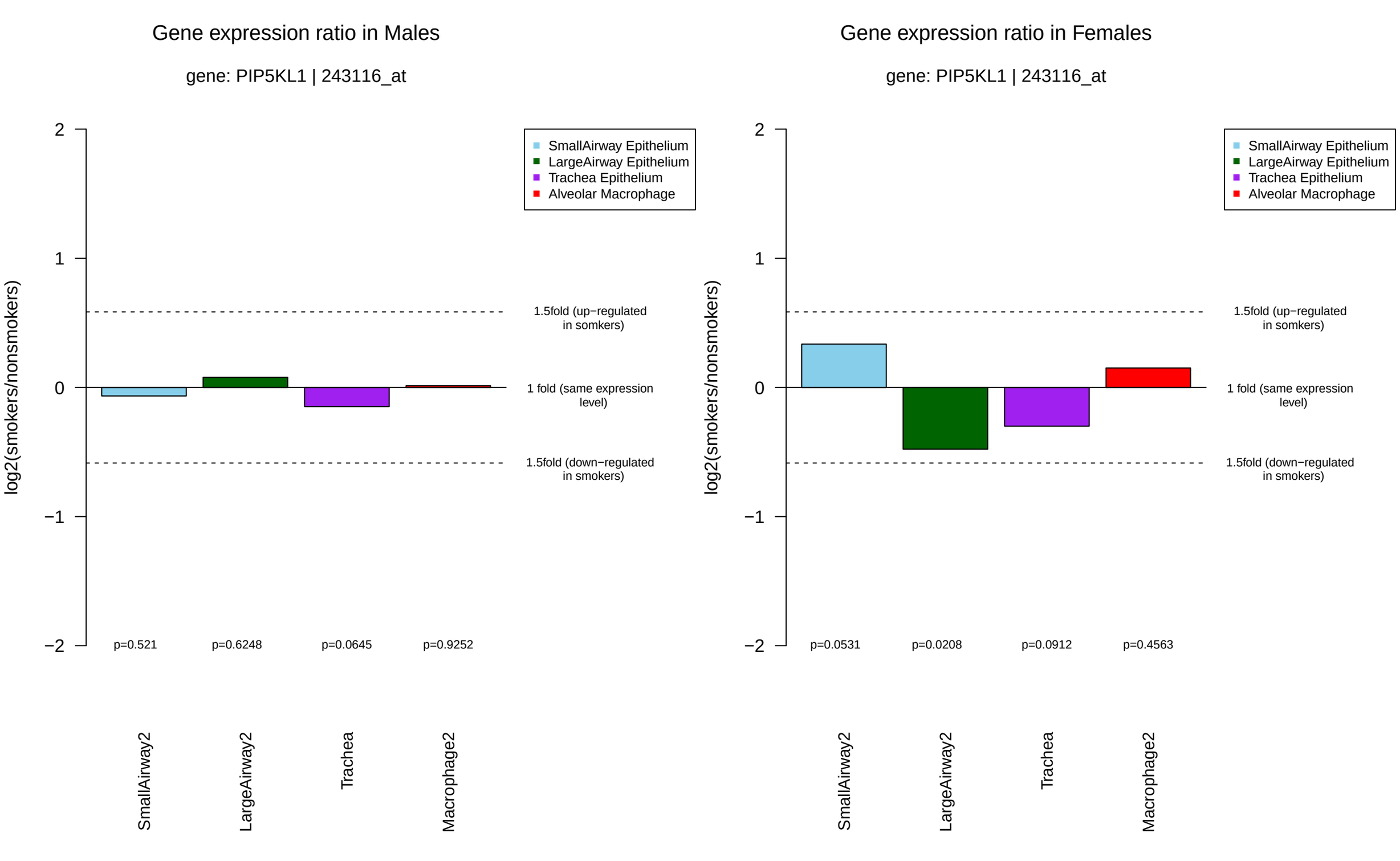
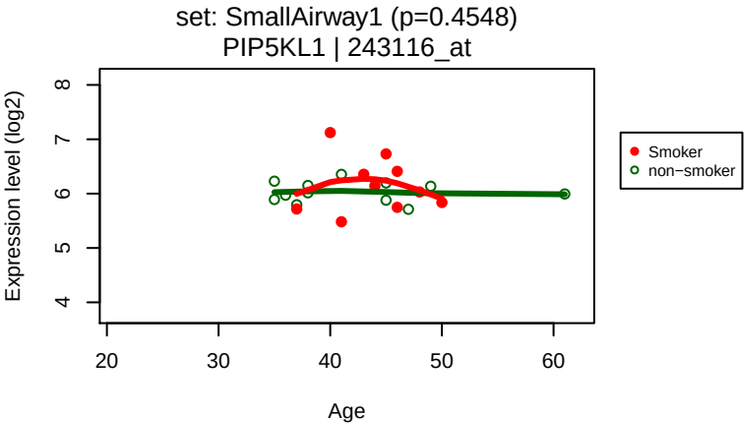
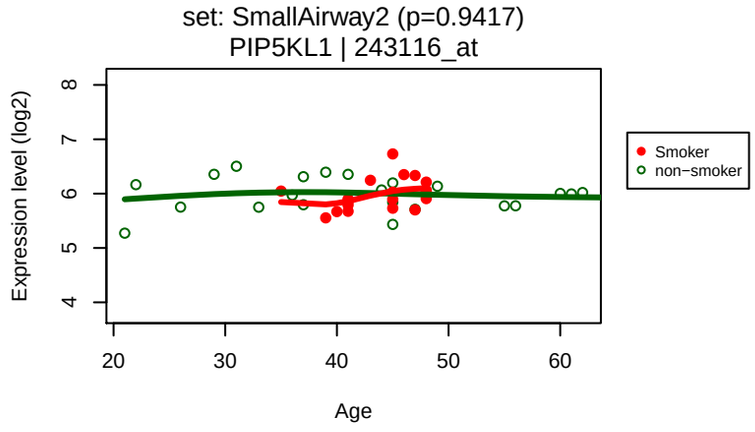
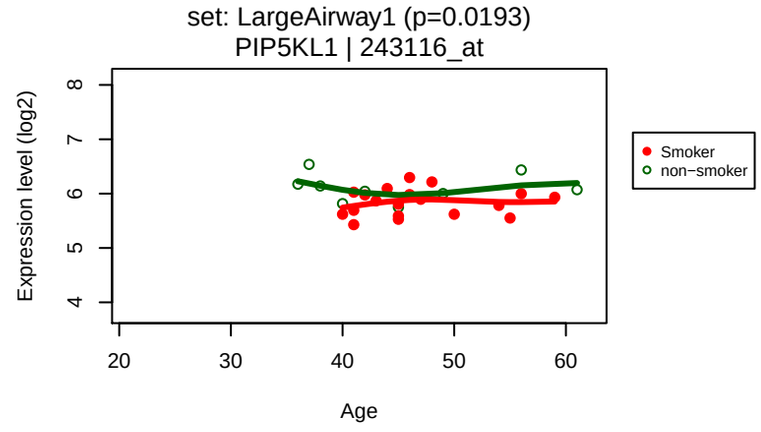

Gene expression of smokers and nonsmokers
gene: PIP5KL1 | 243116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4548
p=0.9417
p=0.7589
p=0.0193
p=0.3618
p=0.003
p=0.285
p=0.5935
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

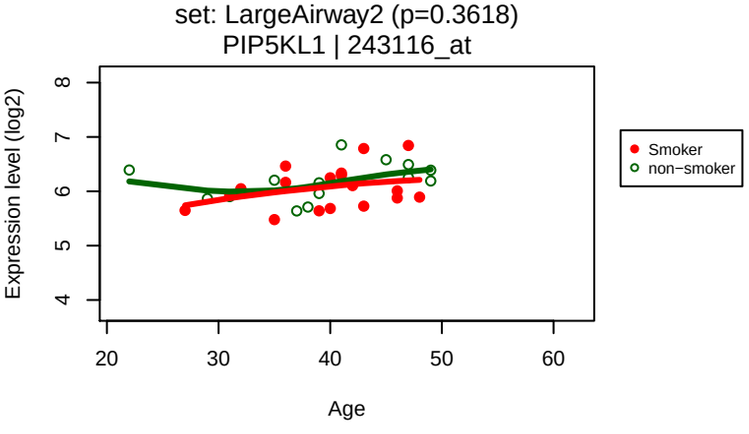

Gene expression ratio (smokers/nonsmokers)
gene: PIP5KL1 | 243116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4548
p=0.9417
p=0.7589
p=0.0193
p=0.3618
p=0.003
p=0.285
p=0.5935
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

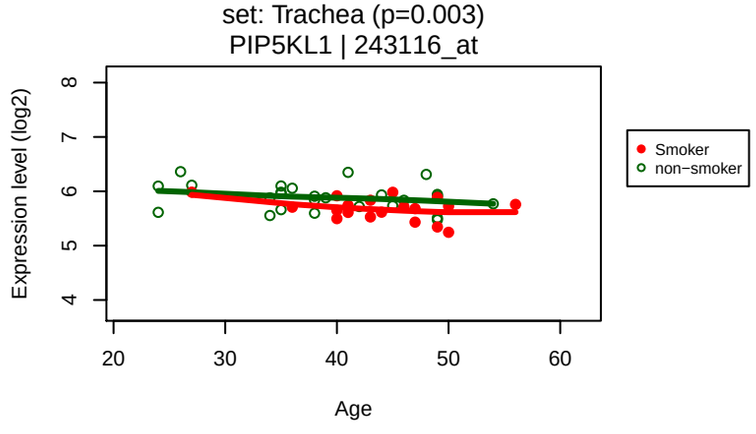

Gene expression ratio in Males
gene: PIP5KL1 | 243116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.521
p=0.6248
p=0.0645
p=0.9252
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIP5KL1 | 243116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0531
p=0.0208
p=0.0912
p=0.4563
SmallAirway2
LargeAirway2
Trachea
Macrophage2

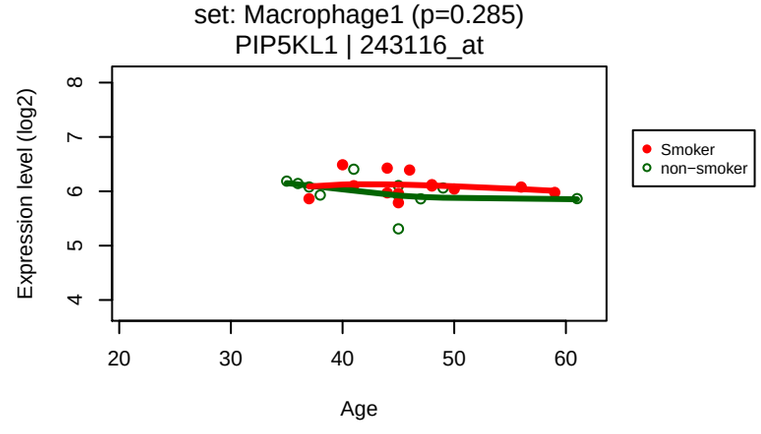

set: SmallAirway1 (p=0.4548)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker

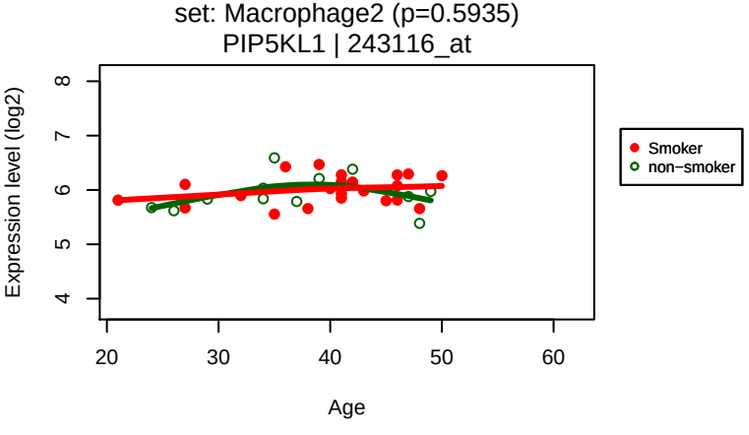

set: SmallAirway2 (p=0.9417)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker

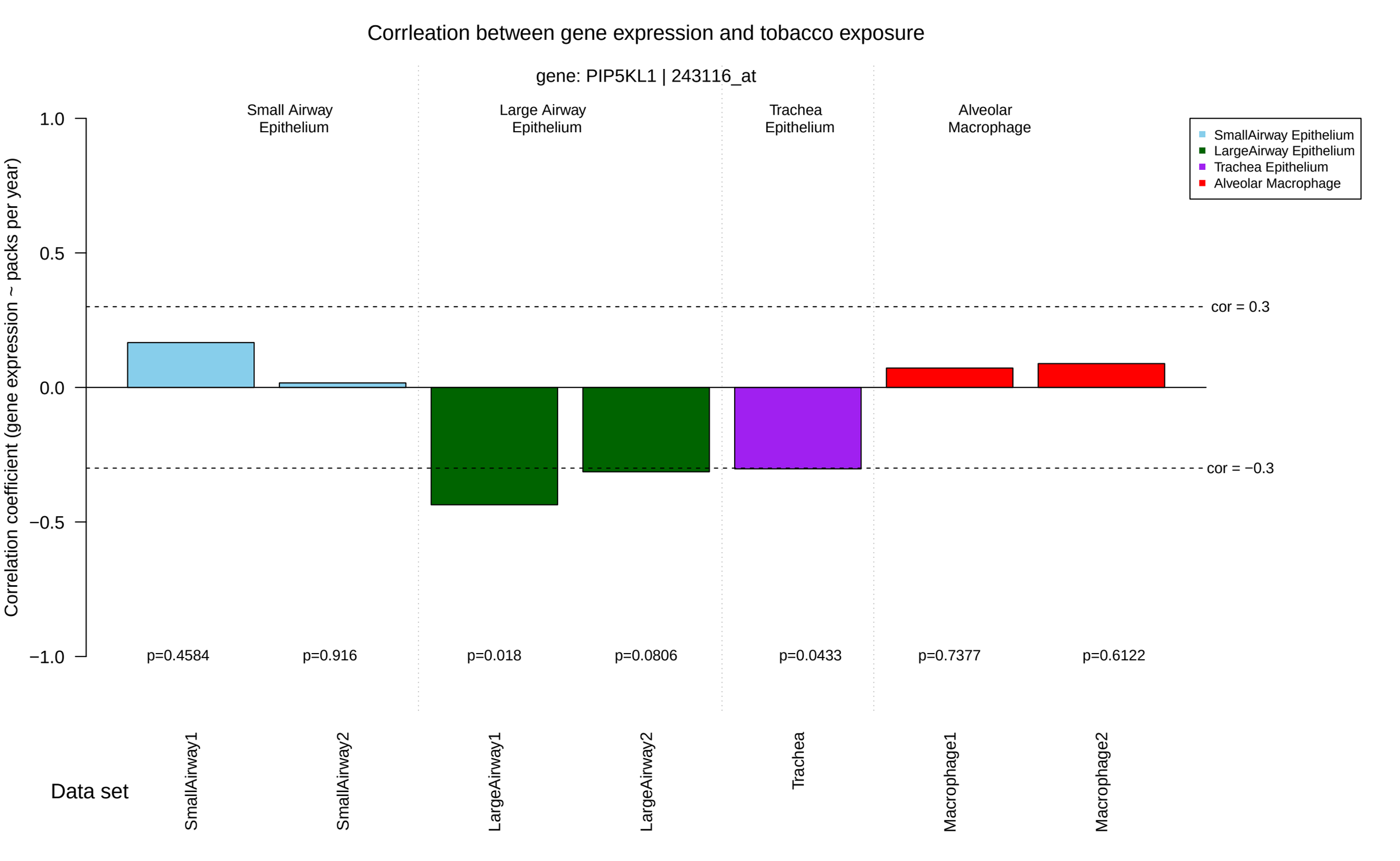

set: LargeAirway1 (p=0.0193)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3618)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.285)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5935)
PIP5KL1 | 243116_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIP5KL1 | 243116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4584
p=0.916
p=0.018
p=0.0806
p=0.0433
p=0.7377
p=0.6122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2